A target-controlled infusion simulator for dexmedetomidine in dogs: validation of the pharmacokinetic model and a proposed correction for anaesthetised dogs

Technical report (simulator version 1.2.1)

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Summary. The simulator calculates the infusion rate of a conventional syringe pump needed to reach and maintain a target plasma concentration of dexmedetomidine in dogs, using the Shafer and Gregg algorithm and the two-compartment model of Levionnois et al. (2022). I compared its predictions with 93 mean concentrations from seven published studies. The model reproduces high concentrations well (2–15 ng/mL) but overpredicts those in the clinical range, particularly in anaesthetised dogs, in which the measured concentration was about half of the predicted one (MDPE −50%). I propose a concentration-dependent clearance, calibrated with the anaesthetised dogs of Pascoe et al. (2006) and Lin et al. (2008), which reduces the bias to −11% in those data and to −3% in conscious dogs not used for calibration. The corrected model does not reproduce the rise in concentration during the first 2–3 hours of anaesthesia. A larger central volume would explain it, but I have not adopted it because it would require loading doses almost three times higher. The model will be tested in a clinical study with pharmacokinetic sampling in 18 anaesthetised dogs.

1. Introduction

In canine anaesthesia, dexmedetomidine is almost always given as a bolus followed by a fixed-rate infusion. With a target-controlled infusion (TCI) system, a plasma concentration is set instead and the program calculates the rate required at each moment, as is already done with propofol in dogs (Beths et al. 2001; Cattai et al. 2019). The simulator I have developed applies this approach to dexmedetomidine. It does not control the pump: it indicates the rate that the anaesthetist should set on a conventional syringe pump and calculates the predicted concentration second by second. For now it is an educational and research tool without clinical validation.

This report summarises how the pharmacokinetic model performs against published data, the correction I propose for anaesthetised dogs and its limitations. It is the starting point of the planned clinical study (protocol TCI-DEX-01).

2. Model and algorithm

The starting point is the final model of Levionnois et al. (2022), obtained in eight conscious Beagles after a 10 µg/kg bolus: V_1 = 1.03 L/kg, V_2 = 0.41 L/kg, systemic clearance Cl = 1.16 L/h/kg and intercompartmental clearance Cl_2 = 1.87 L/h/kg, all scaled linearly with body weight. With these values the program reproduces the published secondary parameters (half-life 0.91 vs 0.92 h; AUC after 10 µg/kg 8.62 vs 8.65 h·ng/mL; MRT 1.24 vs 1.25 h). The amounts in each compartment evolve according to

$$\frac{dA_1}{dt} = R(t) - (k_{10} + k_{12})\,A_1 + k_{21}\,A_2, \qquad \frac{dA_2}{dt} = k_{12}\,A_1 - k_{21}\,A_2, \qquad C_p = A_1/V_1 \tag{1}$$

with $k_{10} = Cl/V_1$, $k_{12} = Cl_2/V_1$, $k_{21} = Cl_2/V_2$ and $R(t)$ the infusion rate.

The algorithm follows Shafer and Gregg (1992) in plasma mode. When the target is increased, the loading dose is infused at a constant rate over 10 minutes, calculated so that the concentration reaches the target exactly at the end; I preferred not to use a rapid bolus in order to limit the hypertensive response. Thereafter, the maintenance rate replaces what is eliminated and what moves to the peripheral compartment. The pump is set in steps of 0.1 mL/h and the rate is only changed when the ideal rate deviates by more than 5% from the programmed one. When the target is lowered or the infusion is paused, the pump stops until the predicted concentration falls to the new value. The program warns for targets above 2.5 ng/mL and blocks those above 4 ng/mL; it limits the rate to 30 µg/kg/h (extending the loading phase if needed) and blocks more than 12 µg/kg in 60 minutes.

3. Validation against published data

I reproduced the protocols of seven studies of intravenous dexmedetomidine in dogs, five in conscious animals (Kuusela et al. 2000; van Oostrom et al. 2011; Honkavaara et al. 2012; Dent et al. 2019; de Sant'Ana Alves et al. 2024) and two in dogs anaesthetised with isoflurane or propofol (Pascoe et al. 2006; Lin et al. 2008), and calculated the concentration predicted by the model at each sampling time. The measured concentrations are group means, taken from the text or digitised from the figures. I digitised the curves of Lin and Pascoe after calibrating the axes with their tick marks, and the resulting means agree with the published ones (in Pascoe, 0.205 and 1.92 ng/mL vs 0.198 and 1.903 ng/mL). Lin et al. dosed by body surface area (25 µg/m² and 25 µg/m²/h), which for their median body weight of 13.7 kg corresponds to about 1.06 µg/kg and 1.06 µg/kg/h.

As a measure of error I used that of Varvel et al. (1992), PE = (measured − predicted)/predicted, with the median (MDPE) as bias and the median of the absolute values (MDAPE) as inaccuracy, applied here to group means. For each study I also estimated the clearance that best reproduces its curve with the volumes and Cl_2 fixed (Table 1).

Overall (n = 93), the MDPE was −41% and the MDAPE 46%, although Lin et al., with 38 points, weigh heavily on this figure. The bias was −31% in conscious dogs and −50% in anaesthetised dogs. The model performs well at high concentrations, as in the 5 µg/kg/h group of van Oostrom et al., and worsens as the concentration decreases. The same is seen in the apparent clearance during the infusions (rate divided by the mean concentration): in conscious dogs it rises from 1.2 L/h/kg at 4.0 ng/mL to 1.9 L/h/kg at 0.53 ng/mL, and in anaesthetised dogs from 1.6 L/h/kg at 1.9 ng/mL to 2.5 L/h/kg at 0.20 ng/mL (Figure 1). In Lin et al., once the dogs were awake, the apparent clearance remained at 2.8 L/h/kg between 3 and 12 h and reached 3.7–4.5 L/h/kg at 24 h, which suggests that it also increases with time. The exception is Dent et al. (2019), with concentrations about four times higher than those of the other studies at similar doses (non-compartmental Cl 0.48 L/h/kg), which I have not been able to explain.

Table 1. Prediction error of the published model in each study.

Study	Administration	State	n	MDPE (%)	MDAPE (%)	Fitted Cl (L/h/kg)
Kuusela et al. 2000	10 µg/kg bolus	conscious	7	−23	31	1.62
	20 µg/kg bolus	conscious	7	−12	21	1.36
Honkavaara et al. 2012	10 µg/kg bolus	conscious	9	+20	20	1.20
de Sant'Ana Alves et al. 2024	2 µg/kg over 2 min	conscious	8	−38	44	3.92
van Oostrom et al. 2011	1 µg/kg + 1 µg/kg/h	conscious	5	−37	37	2.01
	3 µg/kg + 3 µg/kg/h	conscious	5	−18	18	1.62
	5 µg/kg + 5 µg/kg/h	conscious	5	0	15	1.09
Lin et al. 2008	≈1 µg/kg + 1 µg/kg/h, 24 h; isoflurane 2 h	anaesthetised 2 h, then conscious	20	−60	60	3.05
	same, with propofol 2 h	anaesthetised 2 h, then conscious	18	−59	59	3.09
Pascoe et al. 2006	0.5 µg/kg over 6 min + 0.5 µg/kg/h; isoflurane	anaesthetised	4	−52	52	2.51
	3 µg/kg over 6 min + 3 µg/kg/h; isoflurane	anaesthetised	4	−23	23	1.56
Dent et al. 2019	5 µg/kg bolus	conscious	1	+83	83	0.68

n: number of group means. Fitted Cl: clearance that best reproduces the curve with V_1, V_2 and Cl_2 fixed (published Cl: 1.16 L/h/kg). Pascoe et al. measured serum by GC-MS; the other studies measured plasma by GC-MS or liquid chromatography with mass spectrometry.

4. Corrected model for anaesthetised dogs

To correct the bias at low concentrations without changing the behaviour at high concentrations, I made the clearance depend on the plasma concentration C:

$$\mathrm{Cl}(C) = \mathrm{Cl}_\infty + (\mathrm{Cl}_0 - \mathrm{Cl}_\infty)\,\frac{C_{50}}{C_{50} + C} \tag{2}$$

with Cl_∞ fixed at the published value (1.16 L/h/kg) and V_1, V_2 and Cl_2 unchanged. The rationale is that dexmedetomidine reduces cardiac output and hepatic blood flow in a concentration-dependent manner, so that clearance would be higher at low concentrations. Honkavaara et al. (2012) observed something consistent with this: clearance increased when the peripheral effects were blocked with MK-467.

I estimated Cl_0 and C_{50} with anaesthetised dogs only, which is the population of the clinical study: the 20 points from minute 10 onwards of Lin et al. (15–120 min, both groups) and Pascoe et al. (both doses, 1 to 7 h). I minimised the sum of squares of log(measured) − log(predicted), giving each group the same weight (Nelder-Mead from four starting points; integration with LSODA). The initial fit, using the figures given in the text of the papers, gave Cl_0 = 3.15 L/h/kg and C_{50} = 0.92 ng/mL, which are the values used in the simulator. When I repeated it with the complete digitised curves I obtained 3.21 and 0.95, which only changes the maintenance rate at 1 ng/mL from 2.11 to 2.16 µg/kg/h (+2%), so I kept the original values.

Table 2. Error of the published and corrected models (Cl_0 = 3.15 L/h/kg; C_{50} = 0.92 ng/mL).

Data set	n	MDPE published (%)	MDAPE published (%)	MDPE corrected (%)	MDAPE corrected (%)
Anaesthetised, used for calibration	20	−49.9	49.9	−10.7	23.3
Conscious, not used for calibration	63	−37.3	39.5	−2.7	17.8
All, from minute 10 onwards	83	−45.3	45.7	−5.2	18.7

Conscious dogs include those of Lin et al. between 3 and 26 h.

The correction improves most of the studies in conscious dogs that were not part of the fit: Kuusela et al. (MDPE −5% and +4% vs −27% and −15%), van Oostrom et al. at 1 and 3 µg/kg/h (+11% in both, vs −37% and −18%), de Sant'Ana Alves et al. (−32% vs −45%) and Lin et al. from 3 to 26 h (−22% and −19% vs −68% and −63%). After the infusion of Lin et al. was stopped at 24 h, the measured apparent half-life was 0.43 h; the corrected model predicts 0.37 h and the published model 0.85 h. On the other hand, Honkavaara et al. (+24% vs +1%), the 5 µg/kg/h group of van Oostrom et al. (+23% vs 0%) and Dent et al. (+474% vs +83%) become worse (Figure 2).

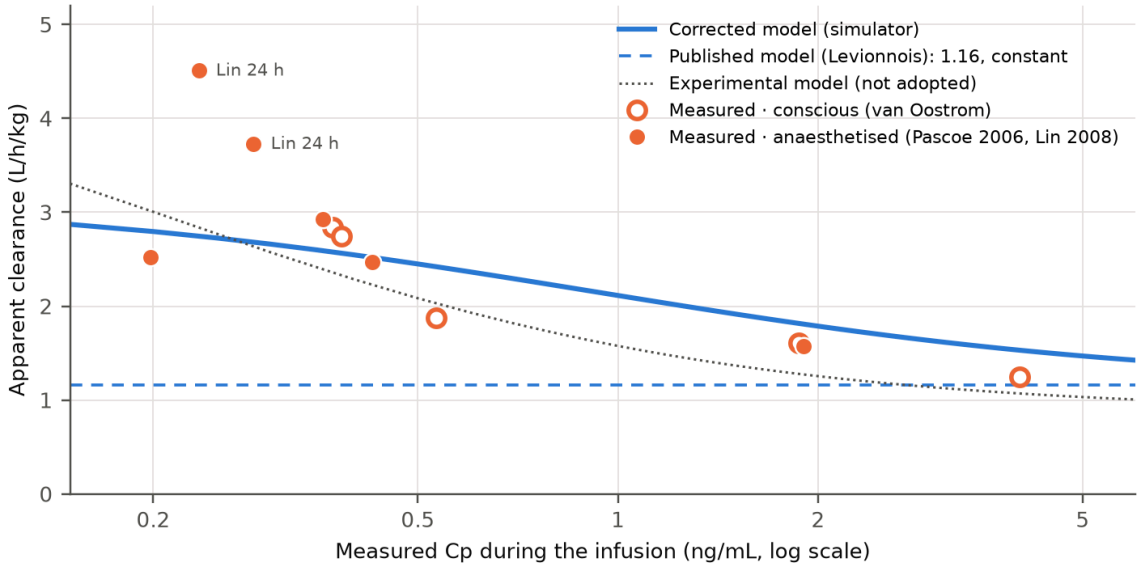


Figure 1. Apparent clearance during the infusion plateau (rate / mean concentration) versus the measured concentration, with the curves of the published and corrected models. The grey dotted line is an experimental model fitted to all studies (Cl_0 4.69; Cl_∞ 0.92; C_{50} 0.20), which did not improve on the published model in cross-validation and which I did not adopt.

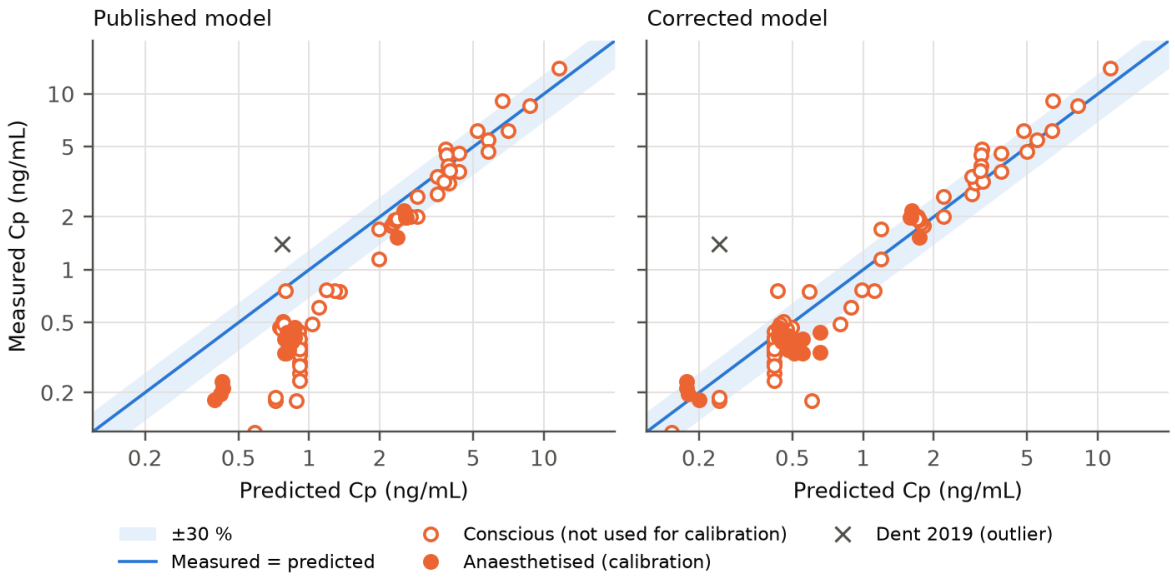


Figure 2. Measured versus predicted concentration with the published model (left) and the corrected model (right). The shaded band indicates ±30%.

With only two calibration studies, cross-validation between them is limited. When calibrated with Pascoe et al. only (Cl_0 3.00; C_{50} 0.55), the model predicts Lin et al. with an MDPE of −32%, compared with −51% for the published model. The reverse is not informative: Lin et al. only have concentrations of about 0.4 ng/mL, so C_{50} is not identifiable and the fit reaches the upper bound allowed, with an error of more than 40% in Pascoe et al. In addition, after a 10 µg/kg bolus the corrected model predicts an AUC of 6.00 h·ng/mL, 31% less than that published by Levionnois et al., although it predicts Kuusela et al. well with the same dose and breed. I interpret this as between-study variability.

The practical consequence of the correction is that the simulator delivers more drug than with the published model, especially at low targets (Table 3). If a dog eliminated the drug as predicted by the published model, a target of 0.5 ng/mL would result in twice that concentration; if it eliminated it like the dogs of Dent et al., about five times. The relative risk is greater at low targets and the absolute risk at high targets. In the clinical study, protection relies on the dose limits, which do not change, on continuous monitoring and on an interim stopping rule.

Table 3. Steady-state maintenance rate and the concentration that would be reached with other clearances.

Target (ng/mL)	Corrected Cl (L/h/kg)	Rate, corrected model (µg/kg/h)	Rate, published model (µg/kg/h)	Cp if Cl = 1.16 (ng/mL)	Cp if Cl = 0.46 (ng/mL)
0.5	2.45	1.22	0.58	1.06	2.66
1	2.11	2.11	1.16	1.82	4.59
2	1.79	3.57	2.32	3.08	7.77
3	1.63	4.88	3.48	4.21	10.61

Cl = 1.16 L/h/kg: published model. Cl = 0.46 L/h/kg: clearance that reproduces Dent et al. (2019).

5. Time course during anaesthesia

The complete curves show something that the corrected model does not reproduce. In both studies the concentration in anaesthetised dogs rises during the first 2–3 h despite the loading dose (in Lin et al., from 0.40 to 0.47 ng/mL with isoflurane and from 0.33 to 0.41 ng/mL with propofol; in Pascoe et al., at

3 µg/kg/h, from 1.52 to 2.0–2.2 ng/mL), whereas the model predicts that it falls after the loading phase. As a result, it overpredicts the concentration at the start and underpredicts it at the plateau. The authors of Pascoe et al. themselves attributed this to an insufficient loading dose.

Using the same data, I also tried to estimate the volumes. With a central volume 2.8 times larger (V_1 ≈ 2.9 L/kg), the model reproduces the initial rise and the plateau of both studies (Figure 3) and the MDAPE falls from 23.7% to 7.3%; increasing V_2 (×4.5) or both volumes gives similar results (8.6% and 8.5%). The model with a larger V_1 also predicts the study it has not seen better (Lin et al., −15% vs −32%; Pascoe et al., +28% vs +43%). However, it is unstable (fitted to Lin et al. only, it gives Cl_0 = 11.8 and C_{50} = 0.05) and performs worse in Lin et al. once the dogs are awake (−40% vs −15%), with a much slower decline after the infusion than measured. It is consistent with a larger central volume during anaesthesia only, but does not prove it.

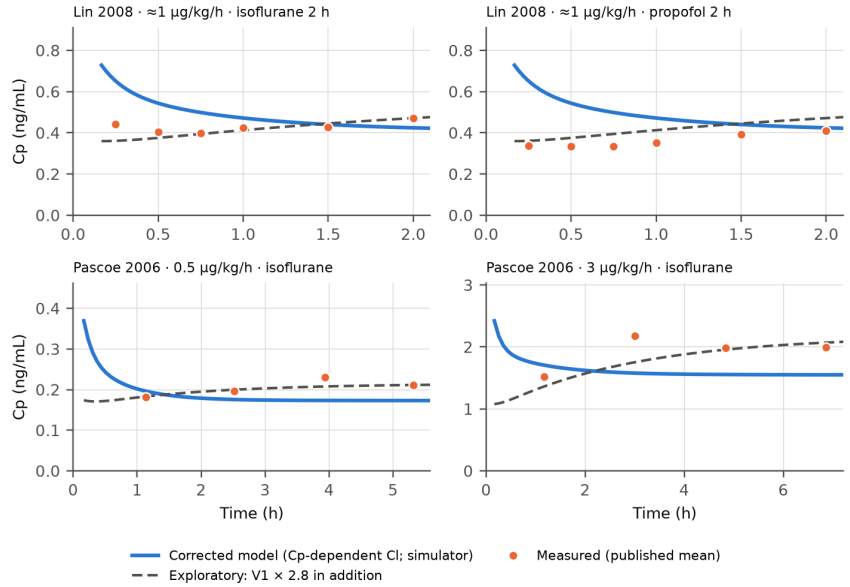


Figure 3. Time course in the anaesthetised dogs of Lin et al. (2008) and Pascoe et al. (2006): published means, corrected model and exploratory model with a central volume 2.8 times larger.

If this interpretation were correct, with the current simulator the concentration would remain below target during the first hour and, at the plateau, 30–40% above target with targets of 0.5–2 ng/mL, because that model gives a lower clearance at 1 ng/mL (1.59 vs 2.11 L/h/kg). I have not adopted it because it would require loading doses almost three times higher (about 2.9 µg/kg in the central compartment at 1 ng/mL and 5.8 µg/kg at 2 ng/mL), with a greater risk of transient hypertension and bradycardia, and because between 0.2 and 2 ng/mL there are no plateau data in anaesthetised dogs to decide between the two models. This is why I added samples at 15 and 20 minutes to the clinical protocol, which should allow V_1 to be estimated.

6. Software verification

Before the validation I checked that the program calculates exactly what its models predict. I wrote a separate reference implementation in Python, with the constants taken from the paper and from the calibration (exact solution by matrix exponential with the published model and an eighth-order Dormand-Prince integrator with the corrected model), and compared it with the simulator in 944 scenarios (472 with each model), with body weights from 2 to 80 kg, targets from 0.25 to 4 ng/mL,

target changes, pauses, manual mode and pump limits. The maximum numerical error was 0.06%; the concentration at the end of the loading phase was within ±1% of the target and the maintenance concentration within ±4.4%. Version 1.2.1 only changes warning texts compared with version 1.2.0 and gives identical results in all comparisons (1844). During the development of version 1.2.0 I found and corrected a defect: with the corrected model and low targets, maintenance deviated by up to 6.7% from the target. I wrote the reference code and the tests myself, with the help of AI tools, so this verification is not independent.

7. Limitations

The main limitation is the scarcity of data in anaesthetised dogs: the correction rests on the group means of two studies (20 points), without individual data. The Varvel error is defined on individuals, and group means hide the variability between dogs. Some of the concentrations were digitised from figures, with a reading error of about ±0.01 ng/mL (±0.02 ng/mL for the propofol points hidden behind other markers in the figure of Lin et al.). The studies used different assays, matrices (plasma or serum) and sampling sites, and in Lin et al. the dose was converted from μg/m². The starting model was obtained in eight Beagles with 120 minutes of sampling, whereas the simulator is used for infusions of up to 240 minutes and in dogs of 2 to 80 kg with linear scaling. The model does not include the increase in clearance over time suggested by Lin et al. at 24 h; within the intended use (up to 4 h) this seems of little relevance, but it has not been verified. Nor is there an effect-site concentration. Finally, I am the author of the model, the analysis and the program, and I have a commercial interest in the simulator.

8. Plan for the clinical study and open questions

Protocol TCI-DEX-01 (version 1.2) includes 30 client-owned dogs (ASA I–II) anaesthetised with isoflurane after premedication with methadone and induction with propofol, 18 of which, weighing 10 to 40 kg, will undergo pharmacokinetic sampling: six for each target of 0.5, 1 and 2 ng/mL, randomly allocated. The pump is guided by the corrected model, in manual mode, for a maximum of 180 minutes. Samples will be taken before the start, at 10, 15, 20, 30 and 60 minutes, at the end of the infusion (or at 120 minutes) and 15 and 60 minutes after stopping it, preferably from an artery, and will be analysed in plasma by a validated LC-MS/MS method (LLOQ ≤ 0.05 ng/mL). For each sample, the concentration predicted by both models with the infusion actually delivered will be calculated. The acceptance criterion will be an MDPE within ±20% and an MDAPE ≤ 30%, as in human TCI. Cl_0, C_{50} and, if possible, V_1 will then be re-estimated with a mixed-effects model. There will be an interim analysis after the first six dogs with samples, and the model will be revised if the measured concentration exceeds the predicted one by more than 50% in two or more dogs. With 18 dogs, the expected half-width of the 95% confidence interval of the MDPE is about ±9 points, assuming a between-dog standard deviation of 20%.

Before starting, I still have several open questions. The main one is structural: whether it is reasonable to attribute the whole difference to clearance or whether the central volume also changes under anaesthesia, and whether a more mechanistic model in which clearance depends on cardiac output would be preferable. I am also concerned that Cl_0 = 3.15 L/h/kg exceeds hepatic blood flow in the dog (approximately 1.8–2.4 L/h/kg of blood), which could point to a blood/plasma ratio different from 1, extrahepatic elimination or a methodological problem in the source studies. Other questions are whether

calibrating with two studies is defensible rather than applying a constant correction factor, whether clearance should be scaled allometrically (exponent 0.75) rather than linearly, and how to treat the data of Dent et al. (2019).

The data (the 99 published means with their sources and the extraction decisions), the analysis scripts and the full reports of the software verification and the external validation are available.

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